

Script.js

```
// -----
```

```
// SMART IRRIGATION
```

```
// -----
```

```
let irrigation = false;
```

```
function toggleIrrigation() {
```

```
    irrigation = !irrigation;
```

```
    const button = document.getElementById("irrigationBtn");
```

```
    const status = document.getElementById("irrigationStatus");
```

```
    if (irrigation) {
```

```
        button.innerText = "Turn OFF Irrigation";
```

```
        status.innerText = "Status: ON 💧";
```

```
        status.style.color = "green";
```

```
    } else {
```

```
        button.innerText = "Turn ON Irrigation";
```

```
        status.innerText = "Status: OFF";
```

```
        status.style.color = "red";
```

```
    }
```

```
}
```

```
// -----
```

```
// START MONITORING
```

```
// -----
```

```
function showAlert() {  
  
    alert(  
        "🌱 AgriSmart Monitoring Started!\n\n" +  
        "IoT sensors are monitoring your farm."  
    );  
}
```

```
// -----  
// CROP IMAGE PREVIEW  
// -----
```

```
document.getElementById("cropImage")  
.addEventListener("change", function(event) {  
  
    const file = event.target.files[0];  
  
    if (file) {  
  
        const image =  
            document.getElementById("preview");  
  
        image.src =  
            URL.createObjectURL(file);  
    }  
});
```

```
// -----  
// AI DISEASE DETECTION  
// -----
```

```
function detectDisease() {

    const result =
        document.getElementById("diseaseResult");

    result.innerHTML = `
        <div class="alert">
            🔍 AI Analysis Completed<br><br>
            🌿 Crop: Healthy<br>
            📊 Confidence: 94%<br>
            ✅ No major disease detected.
        </div>
    `;
}
```

```
// -----
// YIELD PREDICTION
// -----
```

```
function predictYield() {

    const crop =
        document.getElementById("crop").value;

    const area =
        Number(document.getElementById("area").value);

    const yieldPerAcre = {

        Rice: 2.8,
        Wheat: 2.5,
```

```
    Corn: 3.2,  
    Tomato: 8  
};
```

```
const result =  
    area * yieldPerAcre[crop];  
  
document.getElementById("yieldResult")  
.innerHTML =  
    `🌾 Predicted ${crop} Yield:  
    <strong>${result.toFixed(2)} tons</strong>`;
```

```
// -----  
// SIMULATED IoT SENSOR UPDATE  
// -----
```

```
function updateSensors() {  
  
    const temperature =  
        Math.floor(25 + Math.random() * 7);  
  
    const moisture =  
        Math.floor(50 + Math.random() * 25);  
  
    const humidity =  
        Math.floor(60 + Math.random() * 25);  
  
    document.getElementById("temperature")  
        .innerText = temperature + "°C";  
  
    document.getElementById("moisture")
```

```
.innerText = moisture + "%";
```

```
document.getElementById("humidity")
```

```
.innerText = humidity + "%";
```

```
}
```

```
setInterval(updateSensors, 5000);
```

```
// -----
```

```
// YIELD CHART
```

```
// -----
```

```
const ctx =
```

```
document.getElementById("yieldChart");
```

```
new Chart(ctx, {
```

```
type: "bar",
```

```
data: {
```

```
labels: [
```

```
    "Rice",
```

```
    "Wheat",
```

```
    "Corn",
```

```
    "Tomato"
```

```
],
```

```
datasets: [
```

```
{
```

```
    label: "Expected Yield (tons/acre)",
```

```
        data: [
            2.8,
            2.5,
            3.2,
            8
        ]
    }

    ],
},

options: {

    responsive: true,

    plugins: {

        legend: {
            display: true
        }

    }

}

});
```